

ROCKRIDGE POWER & LIGHT COMPANY

Distribution Control Center

Service Interruption Reporting Procedure

RPL-DCC-PRO-003 | Version 4.2

Field	Value
Document ID	RPL-DCC-PRO-003
Title	Service Interruption Reporting Procedure
Version	4.2
Effective Date	2025-02-24
Approved Date	2025-02-19
Owner	Kevin Adeyemi (P23), Manager, Distribution Control Center
Reviewer	Patrick O'Neill (P21), Director, Distribution Operations
Approver	Elena Vasquez (P08), Manager, Regulatory Affairs
Supersedes	4.1 (2024-04-22)
Classification	Internal
Next Review	2026-02-24

Uncontrolled when printed — verify the current version in DCS before use.

Field	Value
Document ID	RPL-DCC-PRO-003
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Effective Date	2025-02-24
Approved Date	2025-02-19
Law As-Of	2024-12-31
Owner	Kevin Adeyemi (P23), Manager, Distribution Control Center
Reviewer	Patrick O'Neill (P21), Director, Distribution Operations
Approver	Elena Vasquez (P08), Manager, Regulatory Affairs
Classification	Internal
Review Cycle	Annual
Supersedes	4.1 (2024-04-22)
Stored In	Document Control System (DCS)

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1. Purpose

This procedure establishes the Distribution Control Center's (DCC) requirements for detecting, classifying, recording, and reporting service interruptions to the Indiana Utility Regulatory Commission (IURC); for compiling and filing annual reliability indices; for issuing advance notice of planned interruptions; and for handing off tree-related outage data to the Vegetation Management Program. It also documents RPL's restoration priorities and the records required to support regulatory compliance.

The DCC operates 24 hours per day, 7 days per week from RPL's headquarters at 400 Wabash Commons Drive, Lafayette, Indiana 47901.

2. Scope

2.1 In Scope

This procedure applies to:

- All sustained service interruptions on RPL's 12.47 kV and 34.5 kV distribution system and 69 kV subtransmission system affecting metered customers;
- Planned interruptions for maintenance, construction, or emergency switching;
- The annual reliability indices report filed with the IURC under 170 IAC 4-1-23(e);
- DCC operators, the DCC Manager (P23), the Director of Distribution Operations (P21), and the Regulatory Affairs team (P08, P09) in their reporting roles.

2.2 Out of Scope

- Momentary interruptions (restorations within 5 minutes or the RPL-defined threshold — see §3.9);
 - Voltage deviations or power quality events that do not result in service interruption;
 - Curtailments under interruptible rate schedules when occurring pursuant to the customer's service agreement (170 IAC 4-1-23(b)(6));
 - Accident reporting (RPL-SAF-PRO-009);
 - FERC/MISO transmission-level events beyond RPL's operated facilities.
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3. Definitions

3.1 Business days — All days other than (A) Saturday, (B) Sunday, or (C) a legal holiday observed by the state of Indiana. (170 IAC 4-1-23(a)(1))

3.2 Nonbusiness days — Saturday, Sunday, or a legal holiday observed by the state of Indiana. (170 IAC 4-1-23(a)(7))

3.3 Customer — For reliability reporting purposes, a metered electrical service point for which an active bill account is established at a specific location. (170 IAC 4-1-23(a)(2))

3.4 Customer of record — Any person, firm, corporation, municipality, or other government agency that has agreed, orally or otherwise, to pay for electric service received from RPL. (170 IAC 4-1-23(a)(4)) For planned interruption notice purposes, RPL uses "customer of record" to identify the party to be notified.

3.5 Interruption — The loss of electrical service to one (1) or more customers connected to the distribution portion of the system. (170 IAC 4-1-23(a)(5))

3.6 Sustained service interruption — A service interruption that is greater than or equal to five (5) minutes unless defined as five (5) minutes or less by the individual utility. (170 IAC 4-1-23(a)(10))

Company practice: RPL applies the statutory five-minute threshold. Interruptions restored within five minutes are classified as momentary and are not counted in SAIFI or SAIDI but are tracked separately.

3.7 Momentary interruption — Company practice: a service interruption that is restored within five (5) minutes of its start. Momentary interruptions are counted in the Momentary Average Interruption Frequency Index (MAIFI) and are not included in SAIFI or SAIDI calculations.

3.8 Planned service interruption (intentional interruption) — A service interruption initiated by RPL to perform scheduled activities, including but not limited to: (A) maintenance; (B) infrastructure improvements; and (C) new construction due to customer growth. Customers of record are typically notified in advance of such events. (170 IAC 4-1-23(a)(8))

3.9 Outage incident — Company practice: a grouping of one or more related outage events occurring on the same circuit or set of adjacent circuits, resulting from the same initiating cause, and managed as a single restoration effort. An incident may include multiple sequential restoration steps, each of which is recorded as a separate event in the OMS.

3.10 Outage event — Company practice: a single device operation resulting in the loss of service to customers. One incident typically comprises multiple events (e.g., a feeder breaker trip followed by sectionalizer isolations and lateral restorations).

3.11 Customer minutes of interruption (CMI) — Company practice: the sum, across all customers affected, of each customer's interruption duration in minutes. For step-restoration incidents, $CMI = \sum (\text{customers restored in step} \times \text{minutes from event start to that step's restoration time})$.

3.12 SAIFI (System Average Interruption Frequency Index) — Calculated by dividing the summation of customers that experienced sustained service interruptions over a specified period of time by the total number of customers served. This index indicates how many sustained service interruptions a customer experiences over a specified period of time. (170 IAC 4-1-23(a)(12))

3.13 SAIDI (System Average Interruption Duration Index) — Calculated by dividing the summation of sustained service interruption durations for a specified period of time by the total number of customers served. This index indicates the total duration of a sustained service interruption for the average customer during a specified period of time. (170 IAC 4-1-23(a)(11))

3.14 CAIDI (Customer Average Interruption Duration Index) — Calculated by dividing the summation of sustained service interruption durations for a specified period of time by the total number of customers interrupted. This index indicates the average time required to restore a sustained service interruption. (170 IAC 4-1-23(a)(3)) $CAIDI = SAIDI / SAIFI$.

3.15 Major event (MED — Major Event Day) — Company practice: a calendar day on which the system-level daily SAIDI exceeds the Major Event Day threshold (TMED). TMED is computed from five years of daily SAIDI data using the IEEE Std 1366™-2012 §2.5β method. Events beginning on a MED are excluded from ex-MED index calculations; all customer-minutes from those events accrue to the start date (begin-

date convention). The definition of major event used for reporting purposes is stated in the annual reliability indices report per 170 IAC 4-1-23(e)(2).

3.16 TMED (Major Event Day threshold) — The daily SAIDI threshold separating routine days from major event days. For 2024 reporting, TMED = 137.19 customer-minutes per customer per day, computed from 2019–2023 daily SAIDI data using the IEEE 1366 §2.5β algorithm. See §15.3 for the worked derivation.

3.17 Investor-owned utility — Any utility that is financed by the sale of securities and whose business operations are overseen by a board representing their shareholders. (170 IAC 4-1-23(a)(6)) RPL is an investor-owned utility under this definition.

3.18 REMC — An electric utility formed under IC 8-1-13. (170 IAC 4-1-23(a)(9)) REMC-specific thresholds in 170 IAC 4-1-23(b)(1)(B) do not apply to RPL.

3.19 State Form 54646 — The IURC-prescribed form "Report of Outage" used to submit initial, update, and final interruption reports. The form captures: utility contact, customers affected and still out, interruption start time, estimated restoration time, location (county/city), cause, and report type.

3.20 Restoration step — Company practice: a documented crew action that restores service to a group of customers. Each step is logged in OMS with the timestamp, customers restored, and the crew lead identifier.

4. Regulatory Basis

ID	Citation	Heading	What It Governs in This Procedure
T4-1	170 IAC 4-1-1	Definitions	Context definitions for "customer" and "disconnection"
T4-2	170 IAC 4-1-2	Applicability of rules	Confirms RPL is subject to this article as an IOU
T4-3	170 IAC 4-1-3	Retention of records	Minimum record retention: at least 3 years for all required records
T4-4	170 IAC 4-1-23(a)	Interruption definitions	SAIFI, SAIDI, CAIDI, sustained/planned interruption, business/nonbusiness days
T4-5	170 IAC 4-1-23(b)	Reportability and reporting intervals	IOU reporting threshold, scheduled intervals, channel, final report trigger
T4-6	170 IAC 4-1-23(c)	Planned interruption notice	Advance notice obligation for interruptions > 1 hour
T4-7	170 IAC 4-1-23(d)	Restoration priorities and emergency procedures	Public health and safety first; written emergency procedures
T4-8	170 IAC 4-1-23(e)	Annual reliability indices report	Annual filing by March 1; SAIDI/SAIFI/CAIDI with and without major events
T4-9	170 IAC 4-1-23(f)	Reliability data retention	7-year retention for CAIDI, SAIDI, SAIFI and supporting data
T4-10	170 IAC 4-1-24	Accident reports	IURC notification of accidents attended with loss of human life (see

			RPL-SAF-PRO-009)
T4-11	170 IAC 4-9-7(g)	Tree-related outage report	Annual tree-related outage report by March 31; content requirements

5. Roles and Responsibilities

5.1 RACI Table

Responsibility	DCC Operator	DCC Shift Supervisor	DCC Manager (P23)	Dir. Dist. Ops (P21)	Reg. Affairs (P08/P09)	Corp. Communications
OMS event creation and cause assignment	R	A	I	I	—	—
Reportability determination	C	R	A	I	I	—
Initial IURC report submission	—	R	A	I	C	—
Update report submissions	—	R	A	I	C	—
Final report submission	—	R	A	I	C	—
Activation of storm mode	—	R	A	A	I	I
Planned interruption customer notice	R	A	I	I	—	—
Monthly reliability index compilation	—	C	R	A	I	—
Annual reliability report filing	—	—	C	C	R	—
Tree-related outage data handoff	—	R	A	I	—	—
Key account notifications (critical customers)	—	R	A	I	—	C

R = Responsible, A = Accountable, C = Consulted, I = Informed

5.2 Role Descriptions

- **DCC Operator** — Monitors OMS alarms, creates and updates outage records, records restoration steps, assigns cause codes, and pages field crews.
- **DCC Shift Supervisor** — Makes reportability determinations, submits IURC reports via email using State Form 54646, activates storm mode, and escalates to the DCC Manager for extended or complex events.
- **Kevin Adeyemi (P23), Manager, Distribution Control Center** — Accountable for all IURC outage reports; reviews initial reports for major events before submission; approves reliability index compilation.

- **Patrick O'Neill (P21), Director, Distribution Operations** — Accountable for storm mode operations, restoration resource deployment, and annual reliability strategy.
- **Elena Vasquez (P08) / Marcus Lee (P09), Regulatory Affairs** — Responsible for compiling and filing the annual reliability indices report with IURC's Electricity Division; coordinates with P21 and P23 on data quality; receives final reports and logs them in RRS-DCC-001.
- **Corporate Communications** — Notified when an event meets the public communications threshold (company practice: incidents affecting $\geq 10,000$ customers or with restoration > 12 hours); issues public updates per RPL-COM-PRO-002.

6. Outage Data Capture in OMS

6.1 Event Creation

The DCC Shift Supervisor creates an OMS incident record within 15 minutes of becoming aware of an outage, using one of the following detection sources:

- **AMI last-gasp** — The AMI head-end (AMI HES) pushes a last-gasp alarm to OMS when a meter loses power. The DCC Operator confirms the alarm against adjacent meter pings before opening an incident.
- **SCADA** — A protective device operation (breaker, recloser) triggers an OMS outage prediction. The DCC Shift Supervisor confirms and extends to affected downstream devices.
- **Customer call** — The DCC Operator opens an outage record from the first call and links subsequent calls from the same geographic cluster to the same incident.
- **Field crew report** — A line crew notifies the DCC of an outage found during patrol. The DCC Operator opens or updates the incident.

Internal performance target: The DCC Shift Supervisor validates the incident type (single_event, storm, or planned) and sets the primary cause category in OMS within 60 minutes of incident creation.

6.2 OMS Incident Fields

Field	Required	Description	Source
`incident_id`	Yes	System-assigned, format `INC-YYYY-nnnnn`	OMS
`incident_type`	Yes	single_event, storm, planned	DCC Operator
`first_start_ts`	Yes	UTC-offset local timestamp of first device operation	OMS / SCADA
`utility_aware_ts`	Yes	Timestamp DCC became aware	DCC Operator
`detection_source`	Yes	ami_last_gasp, scada, customer_call, field	DCC Operator
`peak_customers_out`	Yes	Maximum simultaneous customers affected	OMS auto-compute
`peak_ts`	Yes	Timestamp of peak customers out	OMS auto-compute
`total_customers_affected`	Yes	Unduplicated customer count across all steps	OMS auto-compute
`counties_affected`	Yes	County names (pipe-delimited)	GIS
`critical_facilities_affected`	Yes	Count of critical customers per Key Accounts list	OMS / Key Accounts
`etr_first_ts`	Yes	First estimated time of	DCC Shift Supervisor

		restoration	
`restored_ts`	Yes	Timestamp restoration drops below reportability threshold	DCC Operator
`primary_cause_category`	Yes	Cause from App-D taxonomy	DCC Operator
`cause_code`	Yes	Sub-cause code from App-D	DCC Operator (confirmed within 24 hrs)

6.3 Step Restoration Recording

The DCC Operator records each restoration step in OMS as field crews restore sections of the circuit. Each step entry requires:

- Restoration timestamp
- Number of customers restored in that step
- Cumulative customers still out

Customer-minutes are computed by OMS as: Σ (customers restored in step $\times$ minutes from incident start to that step's restoration time).

6.4 Close-out Verification

The DCC Shift Supervisor verifies incident close-out within 24 hours of restoration:

23. Confirm `restored_ts` is set and all restoration steps are recorded.
24. Confirm `primary_cause_category` and `cause_code` are assigned (not "unknown" unless a patrol has been dispatched and found no cause).
25. Confirm `critical_facilities_affected` is accurate.
26. Confirm `tree_location` is set for all vegetation cause-code events.

Incidents with `cause_code = unknown_patrolled_no_cause` must have a patrol log entry in WAM (work type ENV-INSP or equivalent).

7. Reportability Determination

7.1 Applicability

This section applies to investor-owned utilities only. RPL is an investor-owned utility financed by the sale of securities and overseen by a board representing its shareholders. (170 IAC 4-1-23(a)(6); 170 IAC 4-9-1(a))

7.2 Reporting Threshold — Investor-Owned Utilities

Criterion	Measure	Source Field	Threshold	Must Threshold Be Met?	Citation
Duration	Minutes from `first_start_ts` to `restored_ts`	OMS incident	≥ 120 minutes (2 hours)	Yes — both criteria	170 IAC 4-1-23(b)(1)(A)
Customers affected	`peak_customers_out` at any instant	OMS incident	$\geq 5,000$ customers (lesser of 2% of customers served or 5,000)	Yes — both criteria	170 IAC 4-1-23(b)(1)(A)

Effective customer threshold: RPL serves 407,491 customers as of 2024-12-31. $2\% = 8,149$. The lesser of 8,149 and 5,000 is **5,000**. The applicable threshold is 5,000 customers. (170 IAC 4-1-23(b)(1)(A))

Planned service interruptions are not reportable under 170 IAC 4-1-23(b). The subsection applies only to interruptions that are "not planned."

Interruptible rate exclusion: This subsection does not apply to a curtailment or interruption of service to customers receiving service under interruptible rate classifications when occurring pursuant to the affected retail customer's service agreement. (170 IAC 4-1-23(b)(6))

7.3 Threshold Crossed Timestamp

When the DCC Shift Supervisor determines that both criteria may be met:

27. The DCC Shift Supervisor sets `threshold_crossed_ts` in OMS at the moment both conditions are met. Because peak customers out is typically at or near the start of the incident, and the duration threshold is 120 minutes, `threshold_crossed_ts = first_start_ts + 120 minutes` in most cases.
28. The OMS DCC-ESCALATION paging group is triggered automatically when `peak_customers_out ≥ 5,000`.
29. The DCC Shift Supervisor starts the reporting clock. The initial report is due at the **next scheduled reporting interval** after `threshold_crossed_ts`. (170 IAC 4-1-23(b)(1))

7.4 Determination Workflow

Flowchart TD

```

A[Outage Incident Created in OMS] --> B{incident_type = planned?}
B -- Yes --> C[Not reportable. Record in OMS.]
B -- No --> D{peak_customers_out ≥ 5,000?}
D -- No --> E[Not reportable. Monitor. Re-evaluate if peak rises.]
D -- Yes --> F{Duration ≥ 120 min or likely to reach 120 min?}
F -- No --> G[Not yet reportable. Set timer at threshold_crossed_ts+120.]
F -- Yes --> H[REPORTABLE. Set threshold_crossed_ts. Start reporting clock.]
H --> I[DCC Shift Supervisor submits initial report by next scheduled interval]
  
```

Caution: If an incident begins with < 5,000 customers but subsequently escalates (e.g., a feeder breaker trip following a recloser operation), the DCC Shift Supervisor must re-evaluate reportability as `peak_customers_out` rises.

8. Initial Report

8.1 Timing

The DCC Shift Supervisor submits the initial report to the IURC **by the next regularly scheduled reporting interval** after the incident first crosses the reporting threshold. (170 IAC 4-1-23(b)(1))

The regularly scheduled intervals (Eastern Standard Time — Indianapolis time) are:

- **Business days:** 6:00 a.m., 9:00 a.m., 11:00 a.m., 2:00 p.m., 4:00 p.m., and 9:00 p.m. EST (170 IAC 4-1-23(b)(2)(A))
- **Nonbusiness days (Saturday, Sunday, or Indiana state holiday):** 6:00 a.m., 2:00 p.m., and 9:00 p.m. EST (170 IAC 4-1-23(b)(2)(B))

Example: If `threshold_crossed_ts = 2024-03-15T02:02` (Friday, business day), the initial report deadline is 2024-03-15T06:00.

In cases of extreme emergency, a different reporting schedule may be agreed to by the IURC and RPL until the emergency has ended. (170 IAC 4-1-23(b)(4))

8.2 Channel

Service interruption reports occurring during business days shall be submitted to the IURC and to the Office of the Utility Consumer Counselor (OUCC) via commission-prescribed format. The **preferred method is electronic mail**. Telephone or other types of reports may be made if coordinated in advance with IURC staff. (170 IAC 4-1-23(b)(3))

Internal performance target: The DCC Shift Supervisor sends all reports to both recipients in a single email to the IURC Energy Division and the OUCC, using the State Form 54646 as the email body or attachment.

8.3 State Form 54646 Field Mapping

State Form 54646 Field	RPL OMS / Data Source	170 IAC 4-1-23 Reference
Utility name	Rockridge Power & Light Company	—
IURC utility ID	99012	—
Contact name	Kevin Adeyemi (P23) or on-duty DCC Shift Supervisor	—
Contact phone	1-800-555-0142 (DCC direct)	—
Contact email	kevin.adeyemi@rockridge-pl.example	—
Report type (initial / update / final)	Derived from report sequence	170 IAC 4-1-23(b)(1)
Interruption start time	`first_start_ts`	—
Date/time of this report	`submitted_ts`	—
County/counties affected	`counties_affected`	—
City/municipality	`municipalities_affected`	—
Number of customers affected total	`total_customers_affected`	—
Number of customers still without service	Current count from OMS	—
Estimated restoration time	Current `etr_ts` from OMS	—
Cause of interruption	`cause_reported` (may be "under investigation" initially)	—
Report submitted by	DCC Shift Supervisor name	—

Note: The cause field may read "under investigation" in the initial report. The DCC Shift Supervisor updates the cause in subsequent update reports as field crews identify the cause.

9. Update Reports

9.1 Frequency

The DCC Shift Supervisor submits update reports to the IURC at **each regularly scheduled interval** until electrical service has been restored to the level below the reporting threshold (< 5,000 customers remaining without service). (170 IAC 4-1-23(b)(1))

The intervals are the same as listed in §8.1.

9.2 Content

Each update report via State Form 54646 includes:

- Current `customers_still_out` (updated from OMS)
- Current `etr_ts` (updated if restoration timeline changes)
- Updated cause if field crews have identified the cause

- Whether restoration is below the threshold (if so, the update is also the final report)

9.3 DCC Timer Practice

Internal performance target: The DCC Shift Supervisor sets an OMS reminder timer 15 minutes before each scheduled reporting interval during a reportable event, to ensure the report is compiled and sent on time. The OMS DCC-ESCALATION paging group is automatically re-triggered if a timer expires without a report submission record in OMS.

10. Final Report

10.1 Trigger

The report submitted at the first scheduled interval at which customers still without service have dropped below the reporting threshold (< 5,000 customers) is the **final report** for that interruption period. (170 IAC 4-1-23(b)(1))

10.2 Content

The final report, submitted on State Form 54646 with "Final" marked in the report-type field, contains:

- `customers_still_out_reported` = 0 (or the actual count if below 5,000)
- `etr_reported` = `restored_ts` (actual restoration time)
- `cause_reported` = confirmed cause from OMS
- Notation that full restoration has been achieved

10.3 Post-Final Actions

After the final report is submitted:

30. The DCC Shift Supervisor records the `iurc_final_report_ts` in OMS.
31. The IURC may notify RPL if a written report or further information is required. (170 IAC 4-1-23(b)(5))
32. The DCC Shift Supervisor completes incident close-out per §6.4 within 24 hours.
33. All report records are stored in RRS-DCC-001 in DCS.

11. Restoration Priorities

11.1 Regulatory Priority

RPL shall first attempt to restore service that affects **public health and safety**. (170 IAC 4-1-23(d))

11.2 RPL Restoration Priority Tiers

Tier	Priority Level	Description	Examples
1	Public health and safety	Restoring service to facilities critical to public safety and emergency response	Hospitals, fire stations, police stations, water treatment plants, emergency shelters
2	Critical infrastructure	Facilities whose outage creates imminent public risk	Traffic signals, water pumping stations, nursing homes

3	Subtransmission (69 kV)	Bulk restoration affecting many downstream customers	69 kV line or substation supply failures
4	Substation feeders (backbone)	Primary feeder restoration (12.47 kV / 34.5 kV backbone)	Feeder breaker trips restoring 300–2,500 customers
5	Laterals and fuses	Section isolation and lateral restoration	Sectionalizer and fuse operations restoring 15–300 customers
6	Individual services and transformers	Single-customer or single-transformer outages	Service drops, transformer failures restoring 1–15 customers

11.3 Emergency Procedures

In compliance with 170 IAC 4-1-23(d), RPL maintains written emergency procedures that contain at least:

- 34. **Notification procedures** for emergency response personnel;
- 35. **General location** of equipment, tools, and materials normally needed to restore service;
- 36. **Procedures for notifying** fire, police, medical, and other public officials.

These procedures are maintained in RPL-EMR-PLN-001 (Emergency Response & Storm Restoration Plan). The DCC Shift Supervisor follows RPL-EMR-PLN-001 during any event meeting the plan's activation criteria.

12. Planned (Intentional) Interruptions

12.1 Advance Notice Obligation

Whenever service is intentionally interrupted for any purpose, RPL shall, **except in emergencies**, make reasonable attempts to minimize the inconvenience to affected customers of record. RPL shall make reasonable attempts to notify in advance customers of record whose service is expected to be interrupted for **more than one (1) hour** for scheduled maintenance or facilities upgrades, consistent with safety and security considerations. (170 IAC 4-1-23(c))

This rule does not apply to customer interruptions pursuant to an interruptible tariff or agreement approved by the IURC. (170 IAC 4-1-23(c))

12.2 Notice Process

The DCC Operator or the planning engineer issues advance notice using one or more of the following methods (consistent with the number of customers affected):

- **Robocall or interactive voice response (IVR):** preferred for residential customers; automated from the CIS customer phone number on file.
- **Email:** for commercial and industrial customers where an email address is on file.
- **Telephone call by DCC Operator:** for critical facilities and industrial customers.
- **Door hanger or written notice:** for areas without telephone contact data.

Internal performance target: RPL sends advance notice at least **48 hours** before any planned interruption expected to exceed 1 hour, except in cases listed in §12.3.

12.3 Emergency Exception

The advance notice obligation does not apply when:

- The interruption is required to address an emergency condition (e.g., downed conductor, equipment in imminent failure); or
- Safety considerations make advance notice impracticable.

In such cases, the DCC Shift Supervisor records the emergency/safety basis in the OMS incident record.

12.4 Planned Interruption Records

For each planned interruption, the DCC Operator records the following in the `planned_interruption_notices_2024.csv` log (archived monthly in DCS under RRS-DCC-001):

- Incident ID, notice method, notice sent timestamp, customers noticed, exception code (if applicable), scheduled start time, actual start time.

13. Storm Mode

13.1 Activation Criteria

Internal performance target: The DCC Shift Supervisor activates storm mode when two or more of the following conditions are met:

- ≥ 5 simultaneous open incidents on OMS affecting $\geq 10,000$ customers total;
- Active National Weather Service advisory (tornado watch/warning, severe thunderstorm warning, ice storm advisory) covering ≥ 3 service-territory counties;
- Field crew requests for mutual assistance exceed available internal crew capacity;
- Major natural event (flood, tornado touchdown, widespread ice) is confirmed in service territory.

13.2 Storm Mode Operations

Under storm mode, the DCC Shift Supervisor:

37. Notifies the Director, Distribution Operations (P21) and the DCC Manager (P23) immediately via the OMS DCC-ESCALATION paging group.
38. Activates the Incident Command structure per RPL-EMR-PLN-001.
39. Notifies Corporate Communications per RPL-COM-PRO-002.
40. Maintains reporting cadence per \$8-\$10 — IURC reporting obligations continue unchanged during storm mode.

13.3 Extreme Emergency Reporting Schedule

In the case of an extreme emergency, a different schedule for status reporting may be agreed to by the IURC and RPL until the emergency has ended. (170 IAC 4-1-23(b)(4))

The DCC Manager (P23) requests a modified schedule from the IURC Energy Division (317-232-2785) and confirms any change in writing (email) before deviating from the standard intervals. The agreed schedule is logged in OMS and reported to Elena Vasquez (P08).

13.4 Mutual Assistance

When mutual assistance crews from other utilities are deployed in RPL's service territory, the DCC Shift Supervisor maintains crew tracking in OMS and ensures mutual assistance restoration steps are recorded with the same detail as RPL crew steps.

14. Interruption Records

14.1 Records Table

Record	System	Series ID	Retention	Regulatory Basis
Outage incident records (OMS extract)	OMS / DCS	RRS-DCC-001	At least 3 years (170 IAC 4-1-3)	170 IAC 4-1-3
IURC interruption reports (initial, update, final)	DCS (email archive)	RRS-DCC-001	At least 3 years (170 IAC 4-1-3)	170 IAC 4-1-3
CAIDI, SAIDI, SAIFI indices and supporting workpapers	DCS	RRS-DCC-002	At least 7 years (170 IAC 4-1-23(f))	170 IAC 4-1-23(f)
Daily SAIDI history (for MED/TMED computation)	DCS	RRS-DCC-002	At least 7 years	170 IAC 4-1-23(f)
Planned interruption notice log	DCS	RRS-DCC-001	At least 3 years	170 IAC 4-1-3
Tree-related outage data extract	DCS	RRS-DO-004	At least 3 years	170 IAC 4-1-3; 170 IAC 4-9-7(g)

All records shall be preserved within Indiana at RPL's principal place of business (400 Wabash Commons Drive, Lafayette, Indiana 47901) or at other locations designated after notification to the IURC. Records shall be open for examination by the IURC or its representatives. (170 IAC 4-1-3)

Any retention period set by RPL-LEG-RRS-001 (Records Retention Schedule) that exceeds the statutory minimum governs. Records subject to a legal hold under RPL-LEG-PRO-003 are retained until the hold is released.

15. Annual Reliability Indices

15.1 Filing Requirement

Each investor-owned utility shall file a reliability indices report with the IURC's Electricity Division **on or before March 1** of each year. (170 IAC 4-1-23(e))

- The first report filed under this section shall include data from the **previous three calendar years**. (170 IAC 4-1-23(e))
- Subsequent annual reports shall include data **only from the previous calendar year**. (170 IAC 4-1-23(e))

The report is submitted via the IURC Electronic Filing System (iurc.portal.in.gov) by Elena Vasquez (P08), with Marcus Lee (P09) as alternate, on the form prescribed by the IURC. (170 IAC 4-1-23(e))

15.2 Required Report Content

The annual reliability indices report shall contain: (170 IAC 4-1-23(e))

41. The reliability indices SAIDI, CAIDI, and SAIFI, **with and without major events**, for the RPL system and for each district or region into which its system may be divided;
42. The **definition of major event** used by RPL for reporting purposes;

43. For the reported indices, the **number of customers used for the calculations** and RPL's definition of customer.

15.3 TMED Derivation (Worked Example — 2024 Reporting Year)

RPL uses the IEEE Std 1366™-2012 §2.5β method (the "2.5β TMED method") to identify Major Event Days, as summarized in `corpus/_global/reference/ieee1366_med_method.md`. The threshold (T_MED) is computed from five years of daily SAIDI data preceding the reporting year.

Study period: 2019–2023 daily SAIDI (1,827 daily observations)

Step	Calculation	Value
Non-zero SAIDI days (M)	Count days with daily SAIDI > 0	623 days
μ (mean of $\ln(\text{daily SAIDI})$)	$\sum \ln(s_i) / M$	1.7073
σ (std dev of $\ln(\text{daily SAIDI})$)	$\sqrt{(\sum (\ln(s_i) - \mu)^2 / (M-1))}$	1.1578
T_MED = $\exp(\mu + 2.5\sigma)$	$\exp(1.7073 + 2.5 \times 1.1578)$	137.19 min/customer

Interpretation: Any day in 2024 on which system-level daily SAIDI exceeded 137.19 customer-minutes per customer is classified as a Major Event Day. Customer-minutes from events beginning on a MED are excluded from ex-MED index calculations.

2024 Major Event Days:

Date	Event	Daily SAIDI (min/cust)
2024-03-15	March wind storm	676.85
2024-05-22	May derecho	818.02
2024-07-04	July 4 storm complex	917.32
2024-08-09	August thunderstorm cluster	1,087.16
2024-12-23	December winter storm	989.29

15.4 2024 Annual Reliability Indices — Preliminary (Data Extracted 2025-02-10)

Table 15-A: Annual 2024 Indices

Index	With MED	Without MED
Customer-interruptions (CI)	638,033	445,864
Customer-minutes (CMI)	106,400,234	56,125,189
SAIFI (interruptions/customer)	1.5723	1.0987
SAIDI (min/customer)	262.20	138.31
CAIDI (min/interruption)	166.76	125.88
Major Event Days	5	—
TMED (min/customer)	137.19	—
Customers served	405,805	405,805

Table 15-B: Monthly Reliability Indices — 2024 Ex-MED (Preliminary)

Month	Customers Served	CI (ex-MED)	CMI (ex-MED)	SAIFI (ex-MED)	SAIDI (ex-MED)	CAIDI (ex-MED)	MED Days
Jan	403,997	25,923	3,504,259	0.0642	8.67	135.11	0
Feb	404,305	20,033	2,665,890	0.0495	6.59	133.21	0
Mar	404,593	35,200	4,237,907	0.0870	10.47	120.40	1
Apr	404,900	42,655	7,105,623	0.1053	17.55	166.66	0
May	405,198	54,429	6,513,915	0.1343	16.08	119.70	1
Jun	405,507	55,515	6,070,983	0.1369	14.97	109.36	0
Jul	405,805	64,928	8,544,464	0.1600	21.06	131.60	1
Aug	406,114	31,419	3,245,236	0.0774	7.99	103.24	1
Sep	406,423	35,413	3,319,003	0.0871	8.17	93.76	0
Oct	406,722	30,631	4,669,077	0.0753	11.48	152.45	0
Nov	407,031	18,896	1,702,462	0.0464	4.18	90.14	0
Dec	407,331	30,822	4,546,370	0.0757	11.16	147.44	1
Annual	405,805	445,864	56,125,189	1.0987	138.31	125.88	5

Table 15-C: Vegetation Cause Contribution to 2024 SAIFI/SAIDI (With MED, Preliminary)

Metric	Vegetation Total	% of System (with MED)
CI (customer-interruptions)	213,856	33.5%
CMI (customer-minutes)	40,238,410	37.8%
SAIFI	0.5270	—
SAIDI	99.16 min/customer	—
Inside-ROW CI	127,457	59.6% of veg CI
Outside-ROW CI	44,526	20.8% of veg CI

16. Worst-Performing Circuits Review

16.1 Process

Company practice: The DCC Manager (P23) compiles and presents a worst-performing circuits analysis to the Director, Distribution Operations (P21) quarterly. The analysis uses ex-MED data for the trailing 12 months.

16.2 Top 10 Worst-Performing Circuits — 2024 (Preliminary, Ex-MED)

Note: Circuit rankings are based on customer-interruptions (CI) ex-MED from the 2024 outage events dataset (data extracted 2025-02-10).

Rank	Circuit ID	Service Center	County	CI (ex-MED)	CMI (ex-MED)	Primary Cause	Action Plan Status
1	THT-001-1	THT	Vigo	(from dataset)	(from dataset)	vegetation	In vegetation cycle plan RPL-DO-PLN-002
2	LAF-003-2	LAF	Tippecanoe	(from dataset)	(from dataset)	equipment	Conductor replacement scheduled Q3 2025
3	CRW-005-1	CRW	Montgomery	(from dataset)	(from dataset)	vegetation	Priority trim block 2025-A
4-10	(circuit detail in data/outage_events_2024-01-01_2025-01-31.csv)	—	—	—	—	—	Per quarterly review

Internal performance target: Circuits in the worst-performing top 10 for two consecutive years trigger a root-cause investigation and a corrective action plan filed in WAM.

16.3 Link to Vegetation Management

Circuits with vegetation as the primary cause are referred to the Vegetation Management Program (RPL-DO-PLN-002) for priority trim scheduling. The DCC Manager (P23) provides the monthly extract per \$17 to the Manager, Vegetation Management Program (P22).

17. Tree-Related Outage Data Handoff

17.1 Annual Reporting Obligation

A utility shall file a separate report regarding tree-related outages **by March 31 annually** and whenever the utility makes a change to its vegetation management plan. The report shall include: (170 IAC 4-9-7(g))

- 44. The utility's vegetation management budget;
- 45. Actual expenditures for the prior calendar year;
- 46. The number of customer complaints related to tree trimming;
- 47. The manner in which complaints were addressed or resolved;
- 48. Tree-related outages as a percentage of total outages.

This report is prepared by the Manager, Vegetation Management Program (P22) from the data extract described in §17.2, using the annual filing procedures in RPL-DO-PLN-002.

17.2 Monthly Extract Protocol

- 49. On the first business day of each month, the DCC Shift Supervisor runs the OMS tree-related outage extract (work type DCC - TREE - EXTRACT) covering the prior calendar month.
- 50. The extract includes all events with `cause_category = "vegetation"`, with the following fields: `event_id`, `incident_id`, `start_ts`, `end_ts`, `circuit_id`, `county`, `cause_code`, `tree_location`, `customers_affected`, `customer_minutes`.
- 51. The `tree_location` field is coded:
 - `inside_row` — tree inside the cleared right-of-way;
 - `outside_row` — tree outside the ROW (fall-in);
 - `unknown` — patrol did not determine location.
- 52. The extract is stored in DCS under RRS-DO-004 and emailed to Rachel Stein (P22).

17.3 Monthly Reconciliation

The Manager, Vegetation Management Program (P22) reviews the extract and signs off on the monthly total in WAM (work order type VM-RECONCILE) by the 5th business day of the month. Discrepancies (e.g., events coded as vegetation that were subsequently re-coded) are resolved jointly by P22 and P23 within 10 business days.

18. 2024 Performance Summary (Preliminary — Data Extracted 2025-02-10)

18.1 Overview

All 2024 performance data is preliminary, extracted 2025-02-10. Final 2024 figures will be confirmed in the March 1, 2025 IURC reliability report.

18.2 Reportable Incidents

Table 18-A: 2024 IURC Reportable Incidents

Incident ID	Date	Peak Customers	Duration (hrs)	Counties	Reportable	Initial Report
-------------	------	----------------	----------------	----------	------------	----------------

		Out				Submitted
INC-2024-00001	2024-03-15	26,471	44.5	14	Y	2024-03-15T06:00
INC-2024-00002	2024-05-22	13,742	43.2	14	Y	2024-05-22T06:00
INC-2024-00004	2024-08-09	10,819	58.2	14	Y	2024-08-09T06:00
INC-2024-00005	2024-12-23	7,640	65.1	14	Y	2024-12-23T06:00
INC-2024-02209	2024-04-19	10,580	4.3	Tippecanoe	Y	2024-04-19T09:00
INC-2024-02592	2024-05-13	7,980	4.3	Clinton	Y	2024-05-13T16:00
INC-2024-03211	2024-05-12	9,556	3.4	Tippecanoe	Y	2024-05-12T14:00
INC-2024-04492	2024-07-02	9,732	5.7	Hendricks	Y	2024-07-03T06:00
INC-2024-05377	2024-07-28	11,385	2.8	Hendricks	Y	2024-07-28T14:00
INC-2024-07845	2024-10-12	5,054	5.2	Tippecanoe	Y	2024-10-12T14:00

Total reportable incidents in 2024: 10 | All reports submitted on time: Yes

Report Timeliness: 8 of 10 incidents (80%) submitted in the first 35% of the allowed window. 2 of 10 incidents (20%) used more than 60% of the allowed window (INC-2024-00004 and INC-2024-02592). No late reports.

18.3 Outages and CI by Cause — 2024 Ex-MED (Preliminary)

Table 18-B: 2024 Ex-MED Outages by Cause Category

Cause Category	Events (ex-MED)	% of Total	CI (ex-MED)	CMI (ex-MED)
Vegetation	~2,300	24%	127,000+	~16,000,000
Equipment	~3,000	31%	110,000+	~18,000,000
Animal	~1,400	14%	32,000+	~4,500,000
Weather	~900	9%	28,000+	~4,000,000
Public	~700	7%	24,000+	~3,200,000
Power supply	~200	2%	45,000+	~4,800,000
Operational	~200	2%	10,000+	~1,200,000
Planned	~600	6%	35,000+	~2,500,000
Unknown	~800	8%	35,000+	~1,900,000

Precise counts are in `data/outage\_events\_2024-01-01\_2025-01-31.csv`. Figures above are approximate from preliminary data.

18.4 Monthly SAIFI Profile — 2024 (Preliminary)

Table 18-C: Monthly SAIDI/SAIFI — 2024 With and Without MED

Month	SAIFI (with MED)	SAIFI (ex-MED)	SAIDI (with MED)	SAIDI (ex-MED)
Jan	0.0642	0.0642	8.67	8.67
Feb	0.0495	0.0495	6.59	6.59
Mar	0.3008	0.0870	49.28	10.47
Apr	0.1053	0.1053	17.55	17.55
May	0.2162	0.1343	36.71	16.08
Jun	0.1369	0.1369	14.97	14.97
Jul	0.2095	0.1600	39.49	21.06
Aug	0.1479	0.0774	26.26	7.99
Sep	0.0871	0.0871	8.17	8.17
Oct	0.0753	0.0753	11.48	11.48
Nov	0.0464	0.0464	4.18	4.18
Dec	0.1340	0.0757	38.93	11.16
Annual	1.5723	1.0987	262.20	138.31

Note: The difference between with-MED and ex-MED values in March, May, July, August, and December reflects MED day exclusions. Months with no MED show identical with-MED and ex-MED figures.

18.5 MED Day Summary

Table 18-D: 2024 Major Event Days

MED Date	Incident ID	Primary Cause	Peak CI	SAIDI Contribution (min/cust)
2024-03-15	INC-2024-00001	Vegetation (wind storm)	26,471	676.85
2024-05-22	INC-2024-00002	Vegetation (derecho)	13,742	818.02
2024-07-04	INC-2024-00003	Vegetation (storm complex)	3,529	917.32
2024-08-09	INC-2024-00004	Vegetation (thunderstorm)	10,819	1,087.16
2024-12-23	INC-2024-00005	Vegetation (winter storm)	7,640	989.29

Table 18-E: Indiana IOU Benchmark Comparison (Preliminary)

Metric	RPL 2024 (ex-MED)	Indiana IOU Range (2022 IURC Report)	Within Range?
SAIFI	1.0987	0.71–1.37	Yes
SAIDI	138.31 min/cust	76–178 min/cust	Yes
CAIDI	125.88 min	—	—
MED days	5	3–15	Yes

RPL's 2024 ex-MED SAIDI of 138.31 min/customer places it in the middle-upper range for Indiana investor-owned utilities. The high vegetation contribution (SAIFI 0.527 with MED) reflects RPL's rural overhead exposure across 14 counties and 14,200 overhead circuit miles. The Vegetation Management Program (RPL-DO-PLN-002) targets a 15% reduction in vegetation-related CI over 2025–2026 through accelerated trim cycles on the worst-performing circuits.

19. Training and Drills

Training	Audience	Frequency	ID	Record System	Series
IURC outage reporting — initial qualification	New DCC operators	Once, on-boarding	DCC-T-01	LMS	RRS-TRN-001
IURC outage reporting — annual refresher	All DCC operators and supervisors	Annual	DCC-T-01R	LMS	RRS-TRN-001
Annual IURC reporting drill (tabletop)	DCC Shift Supervisors, P23, P08, P09	Annual (Q4)	DCC-T-02	DCS (drill report)	RRS-TRN-001
OMS cause-code accuracy training	DCC operators	Annual	DCC-T-03	LMS	RRS-TRN-001
Storm mode activation exercise	DCC Shift Supervisors, P21, P23	Annual (Q2)	DCC-T-04	DCS (exercise report)	RRS-TRN-001

The annual reporting drill (DCC-T-02) uses a historical major storm scenario to practice the full reporting sequence from `threshold_crossed_ts` through final report submission, including concurrent update report timers. The drill result is reviewed by P08 to confirm procedure accuracy.

New DCC operators must complete DCC-T-01 and demonstrate proficiency in OMS incident creation, reportability determination, and State Form 54646 submission before being assigned as shift supervisor.

20. Related Documents

Document ID	Title
RPL-DO-PLN-002	Vegetation Management Plan 2025
RPL-SAF-PRO-009	Electrical Accident & Incident Reporting Procedure
RPL-EMR-PLN-001	Emergency Response & Storm Restoration Plan
RPL-COM-PRO-002	Media & Public Communications Procedure
RPL-CMP-REG-001	Regulatory Obligations Register
RPL-REG-CAL-2025	Regulatory Reporting Calendar 2025
RPL-LEG-RRS-001	Records Retention Schedule

21. Revision History

Version	Effective Date	Approved By	Summary of Changes
4.2	2025-02-24	Elena Vasquez (P08)	Incorporated 2024 performance summary (\$18); updated TMED to 137.19 based on 2019–2023 study period; added worst-performing circuits top-10 process (\$16); updated MED dates; added data extract date notation.
4.1	2024-04-22	Elena Vasquez (P08)	Updated customer threshold computation (405,805 denominator); revised App-A example for 2023 storm; added tree-location coding guidance in §17.2.
4.0	2023-05-15	Elena Vasquez (P08)	Full rewrite incorporating storm-mode section (§13); added RACI table; updated TMED calculation to 2018–2022 study period; revised OMS close-out deadline to 24 hours.
3.2	2022-03-07	Thomas Whitfield (P07)	Updated IURC contact information; corrected planned interruption notice threshold reference; minor formatting corrections.

22. Approval Block

Role	Name	Title	Signature	Date
Prepared by (Owner)	Kevin Adeyemi	Manager, Distribution Control Center	_____	2025-02-19
Reviewed by	Patrick O'Neill	Director, Distribution Operations	_____	2025-02-19
Approved by	Elena Vasquez	Manager, Regulatory Affairs	_____	2025-02-19

This document is effective 2025-02-24. Uncontrolled when printed — verify the current version in DCS.

Appendix A — Completed State Form 54646 Example

Incident ID: INC-2024-00001 | **Event date:** 2024-03-15 | **Primary cause:** Vegetation (wind storm)

This appendix shows an example completed State Form 54646 for the 2024-03-15 wind storm (INC-2024-00001). Values correspond exactly to the incident and report-log dataset rows. This was a Major Event Day (SAIDI 676.85 min/customer > TMED 137.19).

A.1 Initial Report — Filed 2024-03-15T02:47

Part A — Utility Identification

Field	Value
Utility Name	Rockridge Power & Light Company
IURC Utility ID	99012
Contact Name	Kevin Adeyemi
Contact Phone	1-800-555-0142
Contact Email	kevin.adeyemi@rockridge-pl.example

Part B — Report Information

Field	Value
Report ID	RPT-2024-00001A
Report Type	☒ Initial ☐ Update ☐ Final
Date/Time of Report	2024-03-15T02:47

Part C — Interruption Details

Field	Value
Interruption Start Date/Time	2024-03-15T00:02
Threshold Crossed Date/Time	2024-03-15T02:02 (start + 120 min)
Counties Affected	Benton, Boone, Carroll, Clinton, Fountain, Hendricks, Montgomery, Parke, Putnam, Tippecanoe, Vermillion, Vigo, Warren, White (all 14 counties)
Total Customers Affected	26,471
Customers Still Without Service	26,471
Estimated Restoration Time	2024-03-15T08:02
Cause	Under investigation (wind/vegetation)

Part D — Submitted By

Field	Value
Name	Kevin Adeyemi
Title	Manager, Distribution Control Center
Date/Time Submitted	2024-03-15T02:47

Submission method: Email to IURC Energy Division and OUCC

A.2 Update Reports Summary

Seven (7) update reports were submitted at each scheduled reporting interval from 2024-03-15T06:00 through 2024-03-16T21:00. Customers still out decreased progressively as field crews completed step-restoration. Cause was updated from "under investigation" to "vegetation/wind" by the second update.

A.3 Final Report — Filed 2024-03-16T21:00

Field	Value
Report ID	RPT-2024-00001Z
Report Type	☐ Initial ☐ Update ☒ Final
Date/Time of Report	2024-03-16T21:00
Customers Still Without Service	0

Actual Restoration Time	2024-03-16T20:32
Cause	Vegetation — wind storm; multiple trees into lines across 14 counties

Appendix B — DCC Reporting Checklist

Use this checklist for each reportable incident. File in DCS under RRS-DCC-001.

Incident ID: _____ | **Shift Supervisor:** _____ | **Date:** _____

Pre-Report Checks

- ☐ OMS incident record created with all required fields (§6.2)
- ☐ incident_type confirmed (not planned)
- ☐ peak_customers_out verified $\geq 5,000$
- ☐ Duration confirmed ≥ 120 minutes (or timer set)
- ☐ threshold_crossed_ts recorded in OMS
- ☐ etr_first_ts set
- ☐ Next reporting interval identified

Initial Report

- ☐ State Form 54646 completed with all fields from §8.3
- ☐ Email sent to IURC Energy Division (cc OUCC) before deadline interval
- ☐ iurc_initial_report_ts recorded in OMS
- ☐ Report saved to DCS, series RRS-DCC-001

Each Update Report

- ☐ customers_still_out_reported updated from current OMS count
- ☐ etr_reported updated if timeline changed
- ☐ Cause updated if field crews have identified cause
- ☐ Email sent before the interval time
- ☐ iurc_update_count incremented in OMS

Final Report

- ☐ customers_still_out_reported = 0 (or $< 5,000$)
 - ☐ etr_reported = actual restored_ts
 - ☐ Cause confirmed in final report
 - ☐ "Final" checked on State Form 54646
 - ☐ iurc_final_report_ts recorded in OMS
 - ☐ Incident close-out per §6.4 completed within 24 hours
-

Appendix C — Notification Contact Sheet

External — IURC

Office	Contact	Phone	Hours
IURC Energy Division (outage reports)	—	317-232-2785	8:15 a.m.–4:45 p.m. ET, Mon–Fri
IURC Electronic Filing System	iurc.portal.in.gov	—	24/7
IURC Consumer Affairs Division	—	1-800-851-4268 (toll-free); 317-232-2712	8:15 a.m.–4:45 p.m. ET, Mon–Fri
OUC (copy on reports)	—	1-888-441-2494	—

IURC mailing address: PNC Center, 101 W. Washington Street, Suite 1500E, Indianapolis, IN 46204.

Internal Escalation

Role	Name	Contact
DCC Manager	Kevin Adeyemi (P23)	kevin.adeyemi@rockridge-pl.example; OMS DCC-ESCALATION
Director, Distribution Operations	Patrick O'Neill (P21)	patrick.oneill@rockridge-pl.example
Manager, Regulatory Affairs	Elena Vasquez (P08)	elena.vasquez@rockridge-pl.example
Regulatory Affairs Analyst	Marcus Lee (P09)	marcus.lee@rockridge-pl.example
Corporate Communications	(role)	OMS DCC-ESCALATION paging group

RPL Customer Outage Line

Customers: 1-800-555-0177 (24/7)

Appendix D — OMS Cause-Code Table

Cause Category	Cause Code	Description	Tree Location Required?
vegetation	tree_inside_row_growth	Tree growth inside ROW contacts conductor	inside_row
vegetation	tree_inside_row_failure	Tree failure inside ROW	inside_row
vegetation	tree_outside_row_fallin	Tree from outside ROW falls into line	outside_row
vegetation	tree_unknown_location	Vegetation cause, location not determined	unknown
weather	wind	Wind without tree contact	n/a
weather	lightning	Direct lightning strike	n/a
weather	ice_snow	Ice loading or snow	n/a
weather	flood	Flood damage	n/a
weather	heat	Heat-related conductor sag or failure	n/a
equipment	oh_conductor	Overhead conductor failure	n/a
equipment	ug_cable	Underground cable failure	n/a
equipment	transformer	Distribution transformer failure	n/a
equipment	cutout_fuse	Cutout or fuse operation (not lightning)	n/a
equipment	arrester	Lightning arrester failure	n/a
equipment	insulator	Insulator failure	n/a
equipment	pole	Pole failure	n/a
equipment	connector	Connector or splice failure	n/a
equipment	recloser_breaker	Recloser or feeder breaker operation	n/a
equipment	substation_equipment	Substation equipment (transformer, bus, relay)	n/a
animal	squirrel	Squirrel contact	n/a
animal	bird	Bird contact	n/a
animal	snake_raccoon_other	Other animal contact	n/a
public	vehicle	Vehicle-into-pole	n/a
public	dig_in	Underground cable damaged by dig-in	n/a
public	vandalism_theft	Vandalism or copper theft	n/a
public	fire	Structure or wildfire	n/a
public	customer_equipment	Customer equipment failure affecting RPL	n/a
public	third_party_contact	Third-party contact with RPL facilities	n/a
power_supply	69kv_line	69 kV subtransmission line fault	n/a
power_supply	transmission_supply_miso	MISO transmission supply interruption	n/a
power_supply	substation_supply	Supply-side substation issue	n/a
operational	overload	Circuit overloaded	n/a
operational	switching_error	Switching error	n/a
operational	protection_miscoordination	Protection mis-coordination	n/a
planned	maintenance	Planned maintenance switching	n/a
planned	construction	Construction or new service	n/a
planned	emergency_switching_for_safety	Emergency switching for safety (treat as planned)	n/a
unknown	unknown_patrolled_no_cause	Patrolled; no cause identified	n/a

Appendix E — Data Dictionary

The following datasets are referenced by this document. Full field-level data dictionaries are in `data/README.md`.

Dataset	File	Rows	Purpose
Outage Incidents	`data/outage_incidents_2024-01-01_2025-01-31.csv`	9,509	One row per incident; includes IURC regulatory columns
Outage Events	`data/outage_events_2024-01-01_2025-01-31.csv`	10,928	One row per device operation / step restoration
IURC Report Log	`data/iurc_report_log_2024-01-01_2025-01-31.csv`	59	One row per IURC report submitted
Planned Notices	`data/planned_interruption_notices_2024.csv`	623	One row per planned interruption; notice compliance
Reliability Indices	`data/reliability_indices_2024.csv`	13	12 months + annual SAIFI/SAIDI/CAIDI with/without MED

Key field cross-references:

- `incident_id` (INC-2024-nnnnn): links incidents to events, report log, and planned notices.
- `event_id` (EVT-2024-nnnnnnn): unique event record within an incident.
- `report_id` (RPT-YYYY-nnnnn{letter}): unique IURC report record.
- `circuit_id` ({SC}-{nnn}-{n}): joins to `circuits_master.csv` for circuit attributes.

Regulatory parameters (from `params` in `scripts/generate\_outage\_data.py`):

Parameter	Value	Source
Customer threshold (IOU)	5,000 (lesser of 2% = 8,149 and 5,000)	170 IAC 4-1-23(b)(1)(A)
Duration threshold	120 minutes (2 hours)	170 IAC 4-1-23(b)(1)(A)
Record retention (interruption records)	At least 3 years	170 IAC 4-1-3
Record retention (reliability data)	At least 7 years	170 IAC 4-1-23(f)
Annual report deadline	March 1	170 IAC 4-1-23(e)
Planned notice trigger	> 1 hour	170 IAC 4-1-23(c)
Tree-related outage report deadline	March 31	170 IAC 4-9-7(g)